

You have 2 hours for completing the exam.

No electronic devices are allowed (i.e. calculators, mobile phones, etc.).

All answers must include the supporting deductions and calculations (if the calculations are too heavy then they can be left implied in the final answer).

If some answer does not include all the necessary information, it will be deemed invalid.

Name and surname:

1. At the Department of Combinatorics there are six researchers, named $A, B, \dots, F$. They are organizing a supper for New Year's Eve, but the secretary of the department is a little bit sloppy and puts the letters of invitation inside the envelopes without looking at the names on each letter and each envelope.

- (a) (1 point) In how many ways is it possible that exactly two letters of invitation end up in the correct envelope?
- (b) (1 point) In how many ways is it possible that at least four letters of invitation end up in the correct envelope?

_____ Solution _____

- (a) If we have exactly two letters in the correct envelope, that means the other four letters are misplaced. First We have to choose the two letters that are in the correct envelope, which can be done in $\binom{6}{2}$ ways. Alternatively, we can choose the four letters that will be misplaced, which can be one in $\binom{6}{4}$ ways. Then we have to multiply this number by the number of derangements of four objects, i.e.

$$\binom{6}{2} \cdot D_4 = \binom{6}{4} \cdot D_4 = 15 \cdot 9 = 135.$$

- (b) In this case there will be four, five, or all six letters in the correct envelope. Following the same reasoning as above, there are

$$\binom{6}{4} \cdot D_2 + \binom{6}{5} \cdot D_1 + \binom{6}{6} \cdot D_0 = 15 \cdot 1 + 6 \cdot 0 + 1 \cdot 1 = 16$$

ways to distribute the letters among the envelopes.

2. (2 points) Prove the following identity:

$$\sum_{k=0}^n \binom{n}{k} k(n-k) 2^{k-1} = n(n-1) 3^{n-2}$$

Hint: Differentiate the binomial formula for $(x+y)^n$ (perhaps more than once) and choose appropriate values for x and y . An alternative is to define a recurrence for the sum and solve it.

_____ Solution _____

Take the binomial formula

$$(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$$

and differentiate both sides, first with respect to x , and then with respect to y , to obtain

$$n(n-1)(x+y)^{n-2} = \sum_{k=0}^n \binom{n}{k} k(n-k) x^{k-1} y^{n-k-1}.$$

Now take $x = 2$ and $y = 1$, and the formula follows.

3. Pell numbers are defined by the recurrence:

$$P_n = \begin{cases} 0 & \text{if } n = 0, \\ 1 & \text{if } n = 1, \\ 2P_{n-1} + P_{n-2} & \text{otherwise.} \end{cases}$$

- (a) (1 point) Find a closed-form formula for P_n (i.e. solve the recurrence) without using generating functions.
- (b) (1 point) Formulate the recurrence in matrix form and find a Cassini-type formula for the sequence $\langle P_n \rangle$.
- (c) (1 point) Find the ordinary generating function $A(z)$ associated with the sequence $\langle P_n \rangle$.

(d) (1 point) Find the radius of convergence of $A(z)$.

_____ Solution _____

(a) This is a homogeneous linear recurrence with constant coefficients.
The characteristic equation associated with the recurrence is

$$x^2 - 2x - 1 = 0,$$

which has two different real roots, namely $1 \pm \sqrt{2}$. Therefore,

$$P_n = \alpha (1 + \sqrt{2})^n + \beta (1 - \sqrt{2})^n,$$

where α and β are constants to be determined. We have

$$P_0 = \alpha + \beta = 0$$

$$P_1 = \alpha(1 + \sqrt{2}) + \beta(1 - \sqrt{2}) = 1.$$

The solution of this linear system is $\alpha = \frac{\sqrt{2}}{4}$ and $\beta = -\frac{\sqrt{2}}{4}$. Hence the closed form of P_n is

$$P_n = \frac{\sqrt{2}}{4} (1 + \sqrt{2})^n - \frac{\sqrt{2}}{4} (1 - \sqrt{2})^n,$$

(b) In matrix form the sequence $\langle P_n \rangle$ is written as

$$\begin{pmatrix} P_{n+1} \\ P_n \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} P_n \\ P_{n-1} \end{pmatrix}.$$

From this we get

$$\begin{aligned} \begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} &= \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^{n-1} \begin{pmatrix} P_2 & P_1 \\ P_1 & P_0 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^{n-1} \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^n \end{aligned}$$

Taking determinants on both sides we get the Cassini-type identity:

$$P_{n+1}P_{n-1} - P_n^2 = (-1)^n$$

(c) The recurrence can be reformulated as

$$P_{n+2} = 2P_{n+1} + P_n.$$

We multiply both sides of the equation by z^n and sum over $n \geq 0$:

$$\begin{aligned}\sum_{n=0}^{\infty} P_{n+2} z^n &= 2 \sum_{n=0}^{\infty} P_{n+1} z^n + \sum_{n=0}^{\infty} P_n z^n \\ \frac{A(z) - P_0 - P_1 z}{z^2} &= 2 \frac{A(z) - P_0}{z} + A(z) \\ \frac{A(z) - z}{z^2} &= \frac{2A(z)}{z} + A(z)\end{aligned}$$

By solving for $A(z)$ we get

$$A(z) = \frac{z}{1 - 2z - z^2}.$$

We may stop here, since we were not asked to decompose $A(z)$ into partial fractions.

- (d) The singularities of $A(z)$ are the roots of the denominator $1 - 2z - z^2$, which are $\sqrt{2} - 1$ and $-\sqrt{2} - 1$. The singularity that is closest to the origin is $\sqrt{2} - 1$, hence the radius of convergence is $|\sqrt{2} - 1| = \sqrt{2} - 1$.

4. Consider the number sequence $\langle a_n \rangle_{n=0}^{\infty} = \langle 1, 0, 3, 0, 5, 0, 7, 0, 9, \dots \rangle$.

- (a) (1 point) Find the ordinary generating function $A(z)$ associated with the sequence $\langle a_n \rangle$.
 (b) (1 point) Find the exponential generating function $B(z)$ associated with $\langle a_n \rangle$.

_____ Solution _____

- (a) The sequence $\langle 1, 0, 3, 0, 5, 0, 7, 0, 9, \dots \rangle$ is derived from the sequence $\langle 1, 2, 3, 4, 5, 6, 7, 8, 9, \dots \rangle$ by taking the even-indexed terms. In the table of ordinary generating functions we see that the generating function of the latter sequence is $\frac{1}{(1-z)^2}$. Therefore,

$$A(z) = \frac{1}{2} \left(\frac{1}{(1-z)^2} + \frac{1}{(1+z)^2} \right) = \frac{z^2 + 1}{z^4 - 2z^2 + 1}$$

- (b) In the table of exponential generating functions we have the sequence $\langle 0, 1, 2, 3, 4, 5, 6, 7, \dots \rangle$, whose exponential generating function is ze^z . The sequence $\langle 1, 2, 3, 4, 5, 6, 7, \dots \rangle$ is derived from the

former by shifting it to the left (or alternatively, by adding the sequence $\langle 1, 1, 1, 1, 1, \dots \rangle$).

$$(ze^z)' = (z + 1)e^z$$

Then we have to take the even-indexed terms and set the odd-indexed terms equal to zero. Hence

$$B(z) = \frac{1}{2} [(z + 1)e^z + (1 - z)e^{-z}] .$$
